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Layer-controlled thinning of black phosphorus by an Ar ion beam†

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Black phosphorus (BP) is one of the most interesting two-dimensional (2D) layered materials due to its unique properties, including a band gap energy change from 0.3 eV (bulk) to 2.0 eV (monolayer) depending on the number of BP layers, for application in nanoelectronic devices. In general, 2D layered materials including BP have limitations in terms of synthesis due to the process factors such as time, temperature, *etc.*, and thus, a thinning technique from the bulk material to a 2D material needs to be used while controlling the removed layer thickness. In this study, layer-controlled thinning of BP was performed by using a controlled Ar^+ ion beam method and the BP thinning characteristics were investigated. By using the near monoenergetic ion energy in the range of 45–48 eV, BP could be thinned with the thinning rate of $\sim$0.55 nm min^{-1} down to bilayer BP without increasing the surface roughness and without changing the chemical binding states. The BP oxide on the pristine BP could also be successfully removed using the same Ar^+ ion beam. 2D BP field-effect transistors (FETs) fabricated with the thinned bilayer–10-layer BPs exhibited electrical characteristics similar to those of pristine BP FETs suggesting no electrical damage on the BP layers thinned by the controlled monoenergetic Ar^+ ion beam.

Introduction

Two-dimensional (2D) materials with a layered structure such as graphene and transition-metal dichalcogenides (TMDs) are considered to be promising layered materials for next-generation electronics and optoelectronics due to their outstanding electrical, optical, and mechanical properties.[1–5] Among the 2D materials, graphene has a high-carrier mobility due to its extremely low-effective mass, but its absence of an intrinsic band gap severely limits its application in semiconductor devices.[6–8] Also, MoS_2 is a representative TMD material which has a band gap depending on the layer thickness and a carrier mobility of 200 $cm^2\ V^{-1}\ s^{-1}$.[9] However, MoS_2 generally shows structural defects such as sulfur vacancies, which will degrade the device performance and limit its applications to electrical and optical devices.[10] Therefore, currently, various other 2D materials are being investigated to fabricate high-performance semiconducting devices.

Black phosphorus (BP) is an emerging 2D layered material and an allotrope of phosphorus which might replace the currently most investigated 2D materials. In particular, bulk BP not only has properties of a direct band gap in comparison with other TMDs such as MoS_2 and WSe_2, but also has a layer-dependent band gap energy from 0.3 eV (bulk) to 2 eV (monolayer).[11,12] These material properties of BP are expected to be beneficial for application in excellent semiconducting devices.[13] To optimize the device properties, it is essential to precisely control the layer thickness of the BP. Currently, 2D BP layers are prepared *via* mechanical exfoliation; however, mechanical exfoliation of BP flakes is not an easy method for obtaining controlled thin 2D BP layers because of the strong inter-layer coupling as Lu *et al.* reported.[14] Therefore, to thin down thick BP flake layers obtained *via* mechanical exfoliation to a few layers of BP, Jia *et al.* studied conventional Ar^+ plasma for controlling the layer thickness of BP.[15] However, this method does not easily control the layer thickness due to the wide range of ion bombardment energy of conventional inductively coupled plasma (ICP) and it can cause possible physical damage on the BP surface due to the high energy tail of the ions. Recently, Lee *et al.* reported a layer-by-layer thinning of BP without structural damage using chemical thinning using a SF_6 plasma.

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In this study, a fluorine residue was left behind on the BP surface after the thinning of BP by the SF_6 plasma, which could chemically react with BP.[16] Therefore, this method may cause partial chemical damage on the BP surface despite the physical damage free process.

Precise control of a few-layer BP film without physical and chemical damage is essential for high-performance electronics. In this study, we investigated a BP thinning method without physical and chemical damage by using a controlled monoenergetic Ar^+ ion beam. After thinning of exfoliated thick BP flakes to a few layers using the monoenergetic Ar^+ ion beam system, the characteristics of the BP layers thinned by the Ar^+ ion beam were compared with those of few-layer pristine BP flakes and the results showed that, by using the controlled monoenergetic Ar^+ ion beam, bulk BP could be successfully thinned to few-layer BP without causing physical, chemical, or electrical damage. In fact, a Ga^+ ion beam used in a focused ion beam (FIB) could also be used instead of the Ar^+ ion beam as investigated in the Ga^+ ion beam patterning of 2D materials such as graphene,[17] but it has been reported that damage occurs during the patterning possibly due to the higher chemical reactivity of Ga compared to Ar. Therefore, even if the Ga^+ ion beam has an energy regime similar to that of the Ar^+ ion beam, the Ar^+ ion beam is more suitable for layer-control and exhibits low-surface damage for various 2D materials including BP.

Experimental details

A bulk BP film with the structure shown in Fig. 1(a) was prepared *via* mechanical exfoliation of bulk BP (ACS Material, LLC) using scotch tape on a SiO_2/Si substrate and the image of the exfoliated BP film was observed using optical microscopy. The thickness of the bulk BP film was determined by atomic force microscopy (AFM; Bruker Innova). Then, the bulk BP film was thinned using a home-made low-energy Ar^+ ion beam system used in this study.

In this ion beam thinning system shown in Fig. 1(b), the energy of the Ar^+ ion beam was varied to optimize the processing conditions to allow low-damage and precise control of the BP thickness. The Ar^+ ion beam source is an ICP type ion gun which is composed of an ICP source for Ar^+ plasma generation and three grids for ion acceleration. 13.56 MHz RF power was applied to the ICP source to extract ions from the plasma source. A positive voltage was applied to the first grid, closest to the plasma source to vary the ion beam energy, and a negative voltage was applied to the second grid to focus the ion beam and to control the flux of the ions extracted from the plasma source while the third grid was grounded. For the thinning of the BP film, 200 W of RF power was applied to the ICP source at 100 sccm of Ar gas flow while the substrate is grounded. A positive voltage of +30 V was applied to the first grid and a negative voltage of −120 V was applied to the second grid. During the thinning, the substrate temperature was kept at room temperature.

The energy of the Ar^+ ion beam extracted from the ICP ion source was analyzed using a retarding grid ion energy analyzer equipped with a current meter (Keithley 2400) and a voltage meter (Hewlett Packard 34401A) positioned at the sample position. The properties of the controlled BP layers were investigated by Raman scattering spectroscopy (WITEC 2000, 532 nm wavelength) and AFM to confirm the thickness of the BP layers and to investigate the possible damage on the BP surface. The chemical binding characteristics of the pristine BP film and controlled thickness of the BP film were investigated by X-ray photoelectron spectroscopy (XPS; ESCA2000, VG Microtech Inc.) using a Mg Kα twin-anode source.

For electrical characterization, back gate BP field effect transistors (FETs) were fabricated on SiO_2 (285 nm)/p^+ doped Si substrates. Before forming BP layers on the substrates, the substrates were cleaned in acetone, isopropyl alcohol, and deionized (DI) water for 10 minutes, sequentially, and dried at 100 °C for 5 minutes. The BP layers were then mechanically exfoliated onto the cleaned substrate. Electron-beam lithography was used to pattern the source/drain electrodes and Ti (10 nm) and Au (50 nm) were sequentially deposited using an e-beam evaporator, followed by a lift-off process. Finally, polymethyl methacrylate (PMMA 950 A4, Microchem) was coated onto the BP FETs to prevent oxidation of the BP layers in the ambient atmosphere. The fabricated back gate BP FETs were characterized by using a semiconductor parameter analyzer (Keithley 4200) at the pressure of 10^{-2} Torr.

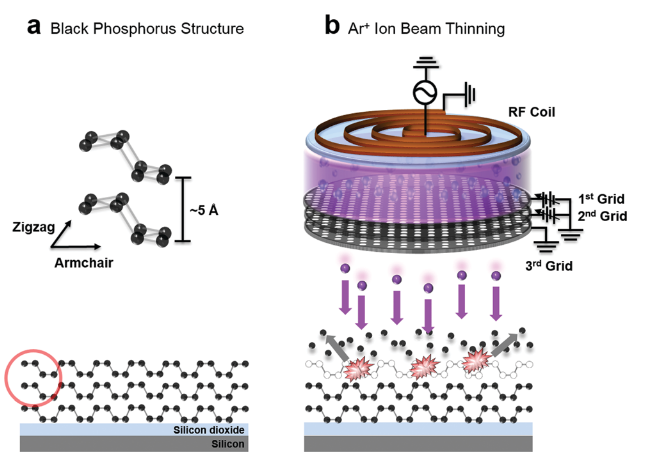

Fig. 1 Schematic diagram of (a) the black phosphorus structure and (b) the Ar^+ ion beam thinning tool used in the experiment.

Results and discussion

In the thinning of black phosphorus using Ar^+ ions, the energy of the ion is very important. If the energy of the Ar^+ ion is lower than the binding energy of BP, BP is not etched. In contrast, if the energy of the Ar^+ ion is too high, the BP underlayers are physically damaged by the Ar^+ ion even though the top BP layer is etched by breaking the bonding of the top BP layer. Therefore, for the controlled thinning of the BP layer without damaging the BP underlayers, adequate Ar^+ ion energy and narrow Ar^+ ion energy distribution are important.[18] Fig. 2(a) shows the Ar^+ ion energy distribution obtained at the substrate position for the

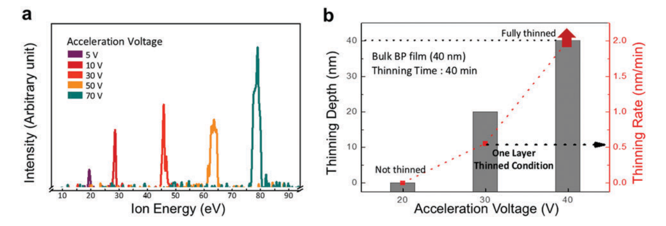

Fig. 2 (a) Ar^+ ion energy distribution measured as a function of acceleration grid voltage of the Ar^+ ion beam source from 5 to 70 V. (b) BP thinning depth and thinning rate measured as a function of the acceleration voltage of the Ar^+ ion beam source for the optimized layer-controlled thinning conditions.

different 1st grid voltages from +5 to +70 V. The energy of the Ar^+ ion extracted from the Ar^+ ion beam source is dependent on the voltage applied to the 1st grid of the three-grid assembly. As shown in the figure, the Ar^+ ion energy distribution for +5 V of the 1st grid voltage was 18–21 eV, 28–31 eV for +10 V, 45–48 eV for +30 V, 62–65 eV for +50 V, and 75–80 eV for +70 V; therefore, the increase of the 1st grid voltage increased the ion energy and the actual peak energy of the Ar^+ ion was higher than the applied voltage due to the existence of plasma potential in the plasma. As the optimum Ar^+ ion energy, +30 V of 1st grid voltage which gives the Ar^+ ions an energy of 45–48 eV was selected for low damage and controlled BP layer thinning conditions. As shown in Fig. 2(b), when +20 V of 1st grid voltage was used, no BP thinning was observed; however, when +40 V of 1st grid voltage was used, an uncontrollably fast etch rate (>2 nm min^{-1}) for BP layer thinning was observed.

BP layers were exfoliated on SiO_2/Si substrates and the thinning of the BP layers was observed using AFM. In general, the thickness of BP layers on SiO_2/Si after the exfoliation is more than 10 layers and they are easily oxidized due to the exposure to water vapor and oxygen in the atmosphere. Therefore, the BP layer thinning experiment was conducted after forming about 10-layer thick non-oxidized BP layers of about 5.8 nm in thickness on the SiO_2/Si substrate which was obtained after the removal of the oxidized and excessive BP layers on the SiO_2/Si substrates using the Ar^+ ions and by the measurement of the remaining BP thickness using AFM (see ESI,† Fig. S1 and S2). Fig. 3(a) and (b) show the optical images of (a) the exfoliated BP layer on SiO_2/Si and (b) the 10-layer BP of $\sim$5.8 nm after the removal of oxidized and excessive BP. AFM surface roughness was measured at the square area in (a) and the thickness of BP during the thinning was measured along the line in (b). Fig. 3(c–f) show the BP layers after the thinning for (c) 2 min, (d) 4 min, (e) 6 min, and (f) 8 min. The BP layers were thinned with the Ar^+ ion energy in the range of 45–48 eV using the Ar^+ ion beam conditions of +30 V of 1st grid voltage and -120 V of 2nd grid voltage while the 3rd grid is grounded (see ESI,† Fig. S3). Ar^+ ions were formed by applying 200 W of RF power to the ICP ion source at 100 sccm of Ar gas flow. When the thickness of the remaining BP layers was measured after the Ar^+ ion beam thinning using AFM, the thickness of the BP layers after thinning for 2, 4, 6, and 8 min was $\sim$4.6, $\sim$3.3, $\sim$2.2, and $\sim$1.0 nm, which correspond to 8, 6, 4, and 2 layers of BP, respectively. Fig. 3(f) shows the remaining BP layer thickness measured as a function of thinning time using the optimized Ar^+ ion beam. As shown in the figure, a straight line was obtained indicating the controlled thinning of the BP layer using the Ar^+ ion beam. The slope was $\sim$0.5 nm min^{-1}, therefore, by using optimized Ar^+ ion beam conditions, the BP layer could be thinned $\sim$1 layer per min. Fig. 3(f) also shows the surface roughness of BP measured by AFM during the thinning. Regardless of the thinning time, the RMS roughness remained at $\sim$0.5 nm, which could be originated from the exfoliated BP. Therefore, no increase of surface roughness was observed in the optimized Ar^+ ion beam thinning.

Using optical microscopy and Raman spectroscopy, the thinned BP layers were observed and the vibrational modes of the BP layer were observed. Fig. 4(a–f) show the optical images observed for various thinned BP layers. Fig. 4(g) shows the Raman shifts of BP layers for the vibrations of the "out of plane mode" (A_g^1) and "in plane mode" (B_{2g} and A_g^2). The A_g^1 peak was observed at 366 cm^{-1}, the B_{2g} peak at 442.3 cm^{-1}, and the A_g^2 peak at 470.3 cm^{-1}.[19] As shown in the figure, up to 4 layers of BP, no peak shifts of the A_g^1, B_{2g}, and A_g^2 modes were observed. However, as the BP layer is decreased from 4 layers to a bilayer, a peak blue shift of the A_g^2

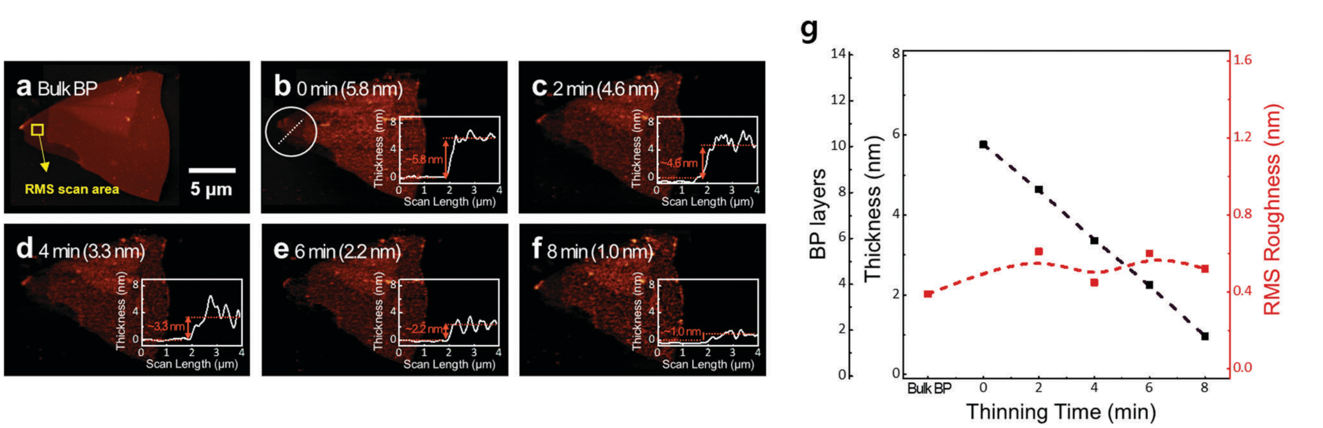

Fig. 3 (a–f) AFM images and the thickness profile of BP layers thinned from bulk to $\sim$1 nm (2 layers) and (g) BP thickness and RMS roughness of BPs thinned with different times (ref. 2, 4, 6, and 8 min).

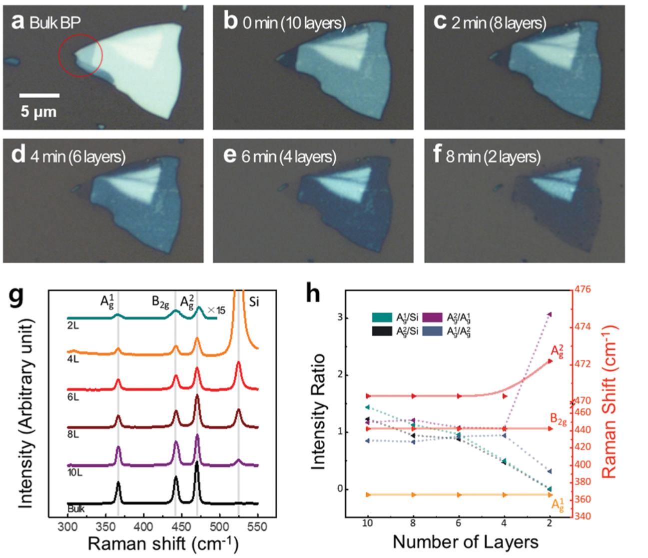

Fig. 4 (a–f) Optical images of BP layers thinned to 2 layers. (g) Raman scattering data of bulk BP and thinned BP with different numbers of layers (bulk, 10, 8, 6, 4, and 2 layers) and (h) Raman shifts of A_g^1, B_{2g} and A_g^2 modes with changing BP layers.

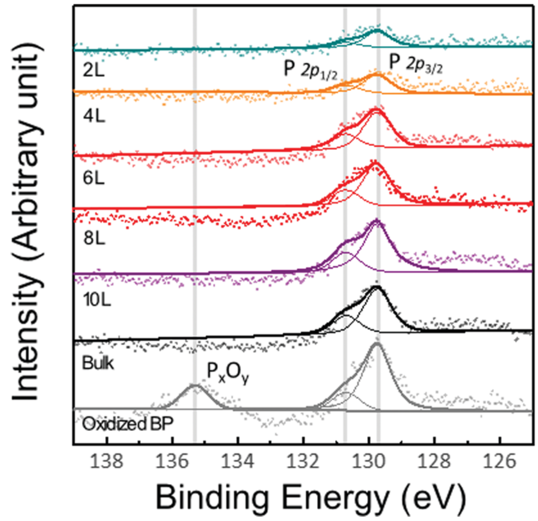

Fig. 5 XPS data of the oxidized BP film exposed in the ambient atmosphere and the layer-controlled BP films (bulk, 10, 8, 6, 4, and 2 layers) with XPS spectra of P $2p_{3/2}$ and P $2p_{1/2}$.

mode from 470.3 cm^{-1} to 472.2 cm^{-1} was observed while showing no peak shifts for the A_g^1 and B_{2g} modes. The peak shift of the in plane A_g^2 mode vibration is known to be from the stiffening of the vibration for the layer thickness thinner than 2 layers, while no changes are observed for the A_g^1 and B_{2g} modes due to the remaining interlayer van der Waals interaction.[20] The Raman shift peak intensities of the A_g^1, B_{2g}, and A_g^2 modes and the intensity ratios of A_g^1/A_g^2 and A_g^2/A_g^1 are shown in Fig. 4(h). In addition, the Raman peak intensity ratios of A_g^1/Si and A_g^2/Si are also shown by using the Raman shift intensity of Si at 524.7 cm^{-1}. The Raman intensity ratios of A_g^1/Si and A_g^2/Si decreased almost linearly with the decrease of the BP layer thickness due to the decrease of BP layers while increasing the Si Raman intensity from the substrate. However, in the case of the intensity ratios of A_g^1/A_g^2 and A_g^2/A_g^1, as shown in Fig. 4(g), the ratios of A_g^1/A_g^2 and A_g^2/A_g^1 remained similar at $\sim$1.0 up to 4 BP layers and, as the BP layer is decreased to a bilayer, the intensity ratios of A_g^1/A_g^2 and A_g^2/A_g^1 decreased to $\sim$0.3 and increased to $\sim$3.0, respectively. (It is known that damage on the BP layer does not change the BP's unique Raman spectrum. Therefore, the changes in the Raman intensity ratios from 4 to 2 layers in Fig. 4(h) are related to the change of the BP layer not to the possible damage on the BP layer by the Ar^+ ion beam.)

The chemical binding states of BP during the Ar^+ ion beam thinning were investigated using XPS and the results are shown in Fig. 5. The XPS spectra were calibrated using the peak at 284.8 eV of the adventitious carbon binding energy peak. BP is easily oxidized in the atmosphere during the transfer of BP layers from bulk BP to SiO_2/Si substrates using an exfoliation method; therefore, the exfoliated BP layers on the SiO_2/Si substrate generally show oxidized BP layers on the surface. In the XPS spectra, on the exfoliated BP layers, a peak at 135.3 eV responsible for P_xO_y was observed in addition to the peaks at 129.8 eV and 130.7 eV related to P $2p_{3/2}$ and P $2p_{1/2}$, respectively. After the Ar^+ ion beam thinning, the peak related to the oxidized BP was removed even after exposure to air to load it into the XPS chamber after the Ar^+ ion beam thinning and no shift of peak intensity peaks related to P $2p_{1/2}$ and P $2p_{3/2}$ and no additional peaks were observed. If the BP layers were damaged during the Ar^+ ion beam thinning process, BP would be rapidly oxidized during the exposure to air and the P_xO_y peak will emerge. Therefore, using the optimized Ar^+ ion beam thinning, controlled BP thinning to 2 layers can be achieved without damaging the BP layers.

To investigate the effect of Ar^+ ion beam thinning on the electrical properties of the BP layers, a BP FET device with 10 BP layers was fabricated and its electrical characteristics were investigated while thinning the BP layers in the BP FETs from 10 layers to a bilayer by using the optimized Ar^+ ion beam. As shown in Fig. 6(a), the back gate FETs with 10 BP layers were fabricated on the SiO_2/Si substrate and the electrical characterization of the FETs of different BP layers was achieved by repeating electrical measurements before and after the BP layer thinning process using the Ar^+ ion beam. The characterization process was performed in an oxygen free atmosphere to minimize the surface degradation of the BP during the measurement. Fig. 6(b) shows representative drain current (I_D)–gate voltage (V_G) characteristics of layer-controlled BP FETs at a drain voltage of 100 mV, which exhibit p-type semiconductor behaviors. Fig. 6(c) shows the statistical field-effect mobility and the on–off current ratio of the FETs as a function of the number of BP layers. The field-effect mobility was calculated using the following eqn (1)

$$\mu = (L/WC_{ox}V_D)\cdot(dI_D/dV_G) \quad (1)$$

where μ is the field-effect mobility, $L/W = 2$ is the channel dimension, and $C_{ox} = 1.2 \times 10^{-4}$ Fm^{-2} is the gate capacitance. The field-effect mobility values were estimated to be 19.2–10.2 cm^2 V^{-1} s^{-1} as the BP layer was decreased from 10 layers to a bilayer. As the BP

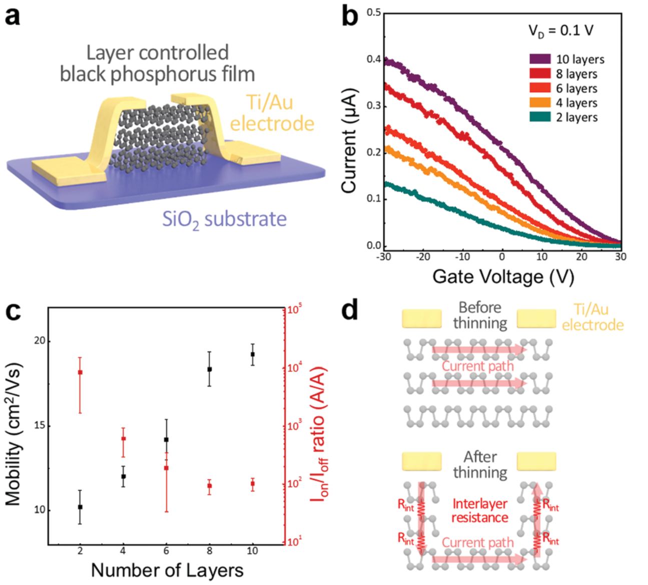

Fig. 6 (a) Schematic diagram of fabricated BP FET. (b) Drain current (I_D)–gate voltage (V_G) transfer characteristics of back-gated BP FET devices (10, 8, 6, 4, and 2 layers thinned using the BP FET devices with 10-layer BP). (c) The current on–off ratio and field-effect mobility as a function of number of BP layers. (d) Schematic illustrations for the BP FET before and after thinning using the Ar^+ ion beam technique with a resistor network model.

layer thickness decreases, the I_D and μ values decrease, which is consistent with previous results[15] reported for BP FETs with the BP layer thickness below 10 nm. This dependence of I_D and μ on the BP thickness is because the charged impurity scattering is more severe for thinner layers.[21,22] In our device, as shown in Fig. 6(d), additional access resistance (R_{int}) exists between the electrode contact and the channel and it also increases as the BP layers are thinned from 10 to 2 layers. However, as shown in Fig. 6(c), the current on–off ratio (I_{on}/I_{off}) increased with the decrease of BP layers in the BP FET and a high I_{on}/I_{off} of 7000 was observed for the bilayer BP FET. The increase of the current on–off ratio with decreasing BP layer thickness shows that the band gap increases due to the quantum confinement effect found in layered materials,[23,24] and it confirms that the BP layer in the BP FET has been thinned to a bilayer controllably without damaging the BP layer surface by the Ar^+ ion beam technique.

Conclusions

We have demonstrated the layer-controlled thinning of BP using an optimized monoenergetic Ar^+ ion beam with ion energy in the range of 45–48 eV, which showed different thicknesses of BP depending on thinning time. BP layers could be thinned with the thinning rate of $\sim$1 layer per min down to bilayer BP without physically and chemically damaging the BP layer surface. 2D BP FETs fabricated with 10-layer BP and thinned down to bilayer BP exhibited electrical characteristics of a high-drain current and 7000 of I_{on}/I_{off} ratio when the device was thinned down to bilayer BP indicating no electrical damage on the BP layer by the optimized monoenergetic Ar^+ ion beam. It is believed that the Ar^+ ion beam technique can also be used to control the thickness of other 2D materials in addition to BP without any damage on the surface and it can be a facile and promising thinning method for fabricating various other 2D-based devices.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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